



Flash Flood Forecasting Based-EF5 Model using Distinct Interpolation Methods: An Ensemble Framework

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Abstract

Predicting runoff plays a key role in flood warning systems in regions like Kendrapara district, where major rivers such as the Mahanadi, Brahmani, and Baitarani meet. These areas pose difficulties to develop a precise flood forecasting model because of their mountainous landscape and frequent flash floods. The Ensemble Framework (EF5) for flash flood forecasting shows promise as a technique without satellite rainfall data. This research applied the EF5 model to build a flood forecast system for the Kendrapara district, focusing on zones prone to recurrent flash floods. Several benchmark machine learning models namely Support Vector Machine (SVM), Random Forest (RF), Gradient boosting machines (GBM) and Long Short-Term Memory Network (LSTM) are also applied for proving effectiveness of the EF5 model. Results confirmed that the EF5 model performed well during validation with the Nash-Sutcliffe efficiency (NSE) and Pearson's correlation coefficient (PCC) routinely exceeding 0.82. Notably, spline interpolation yields the best results, with PCC and NSE values above 0.9, followed by Inverse Distance Weighting (IDW), which comes in second place with PCC and NSE of approximately 0.8; Kriging performs the worst, with PCC and NSE of approximately 0.4 to 0.6. Nonetheless, disparities in efficiency are observed across several interpolation techniques. Furthermore, lead and lag lengths have a considerable impact on the model's predicted accuracy. Shorter lead times correspond to greater PCC and NSE values, which surpass 0.8. However, as the projected lead time rises, these values rapidly decrease, nearing zero or even becoming negative. Given the EF5 framework's simplicity and validity in Kendrapara's tiny mountainous basins, further study and verification are required to improve flood prediction skills in this region.

Keywords Flood forecasting · Forecast lead time · Hydrological model · Rainfall interpolation

1 Introduction

Amongst the most catastrophic natural hazards worldwide are floods. In the last decade, flood events have impacted over 2 billion people worldwide, accounting for over half of all natural disaster related deaths (Alderman et al. 2012; Samela et al. 2016; Le et al. 2021). An abrupt water rush into a narrow canal or over a slanting surface triggered by an unexpected, heavy precipitation event in the upstream hilly regions is commonly referred to as a flash flood (Roy et al. 2019; Jhong et al. 2024; Zhou et al. 2024; Zhai et al. 2025). Because of their quick and erratic rise inflow rate and water levels, flash flood events have attracted substantial consideration from disaster management officials and riverine inhabitants in hilly areas (Bisht et al. 2018; Nagamani et al. 2024). Flash floods are distinct from other flood disasters in that they occur suddenly and quickly (Choudhury and Haque 2016; Das et al. 2020; Majeed et al. 2023; Ikeuchi et al. 2025). The widespread adoption of traditional flood protection structural measures was lowered by the increased frequency of catastrophic flood events driven by climate change (Shah et al. 2018; Hadi et al. 2020; Kuang and Liao 2020). Globally, the severity and rate of flash floods are increasing due to the intricate consequences of climate change, which have exacerbated extreme rainfall events (Xiong et al. 2024). Because of this, it is now more important than ever to establish efficient hazard prevention and mitigation strategies (Hu et al. 2024; Shi et al. 2024; Charpentier-Noyer et al. 2025). One common non-structural measure for forecasting flash floods and their early warning systems is realtime rainfall runoff modeling (Xu et al. 2024). Many studies and practices consider these models to be among the most accurate and dependable flood forecasting techniques because they can depict runoff processes with highly time varying and nonlinear properties (Sun et al. 2018). Since flash floods are more damaging and differ hydrologically from those in vast basins, the scientific community is becoming more interested in forecasting and warning about them. Since 2000, floods have claimed thousands of lives in India; majority of those deaths were brought on by flash floods. The majority of those flash floods have happened in hilly, unprotected regions thus far. The absence of hydrological data makes forecasting and flash flood warnings extremely challenging. Extreme flow into a region that is typically dry or a sudden rise in water level above a flood threshold are characteristics of flash floods. Though the exact timing varies globally, flash flood occurrences usually occur minutes to hours following the precipitation event that caused them. Numerous research studies have been conducted on runoff prediction models, including data driven (Moosavi et al. 2022; Yaseen 2023). Two common multiple satellite rainfall products that are reasonably good at characterizing rainfall trends are GPM and TRMM (Feng et al. 2024).

An uncalibrated hydrological model was used as part of EU-FLASH project for simulating flow events in a Mediterranean catchment (27 km²) in Israel and analyzing various factors impacting flows (Rozalis et al. 2010). Available data from GIS, maps, and field investigations were utilized for defining model constraints, and no further calibration was performed for obtaining a better fit amid calculated and actual flow data. They found that the applied model better predicted flash floods produced by short-lived, intense convective storms. In contrast, model performances for moderate and low flows produced by more extensive winter storms were relatively poor. Chen et al. (2013) coupled Ensemble Square Root-Filter (EnSRF) with conceptual rainfall runoff model HyMOD and applied it to two small catchments in China and America susceptible to flash flooding.

Their study revealed the efficiency and benefits of employing data integration into a hydrological model for improving flash flood forecasts with probable application to a real time alert system. The flash flood generating storms and subsequent hydrological responses in watersheds in Pyrenean foothills up to Aude province were studied, selecting small to medium sized watersheds with contrasting properties (Garambois et al. 2014). They found that flash floods in the Pyrenean watersheds were generated by precipitation near watershed outlet and at lower topography. The sensitivity of streamflow simulation was evaluated in urban watersheds for hydrological modelling to estimate flash floods and spatiotemporal resolution of precipitation input in highly populated Dallas Fort Worth Metroplex (DFW) (Rafieeinassab et al. 2015). Their work used spatiotemporal resolutions for the hydrologic modeling system applied over Grand Prairie, Fort Worth, and Arlington cities in DFW. HiRes Flood UCI was developed by integrating a hydraulic model (BreZo) with NWS's HL-RDHM for modeling flash floods at decameter resolution (Nguyen et al. 2016). The HL-RDHM is utilised as a rainfall runoff generator in the integrated model, replacing its routing scheme, and together with BreZo, for predicting localised flood velocities and depths. The obtained outcomes from the applied model with optimised parameters demonstrated probable utility of the coupled model for estimating heavy rain-persuaded flash floods over Korean Peninsula. Warnings of flash floods were forecasted and provided in a timely and accurate manner, using a newly developed distributed model (China flash flood hydrological model, CNFF-HM), considering data from Zherong and Shunchang provinces (Wang et al. 2017). Coupled modeling (HEC-RAS and HEC-HMS) was investigated in a small urban channel in Campo Grande, Mato Grosso do Sul, utilising high resolution data on flood occurrences (Bruno et al. 2022). Results indicated that the channel has a natural propensity to flood in particular sections, which become more negotiated because of local conditions and landuse changes. A methodology for flash flood Early Warning Systems and a combined hydrometeorological modeling framework were proposed in Chugoku province, Japan (Trošelj et al. 2023). Their findings indicated that proposed combined approach allows for real time precise flash flood forecasts in the region. A Frequentist method combined with Flash Flood Guidance (FFG) was proposed for presenting the flash flood occurrence probability based on historical precipitation events in Posina River basin in Italy (Wu et al. 2023). The use of the methodology in the basin demonstrates that decision making for flood forecasting is easier with developed methodology than with traditional FFG approach. Flash flood hazards in Al-Rashrash and An-Nawayah drainage basins in southeastern Cairo, Egypt, were assessed and predicted by computing peak flow, discharge volume, velocity, and flood depth using HEC-RAS and HEC-HMS software (El-Bagoury and Gad 2024). They developed a system which utilizes H-Q equation and rainfall runoff inundation (RRI) model which can predict precisely when water levels of the specified rivers will reach Flood Risk Level (Ikeuchi et al. 2025). This system was employed in 200 rivers in Japan with varied geological characteristics and diverse rainfall, and was verified during flood season. Their outcomes showed that applied system showed good potential in enhancing and strengthening flood warning and evacuation systems. The application of various machine learning models for predicting flash flood is provided in Table 1.

The EF5 model in Flamig's study was created to use high resolution precipitation forcing records for food forecasting. However, such rainfall input circumstances are challenging for small watersheds in hilly regions. These precipitation products from

Table 1 Literature review of different applications for flash flood

Authors and year of publication	Methods used	Datasets	Performance metrics	Key findings
(Zhou et al. 2025)	CNN, LSTM, Transformer	44 flood events	NSE, R, PE, TE	Transformer model exhibited significant potential for longer forecast correction times
(Soares et al. 2025)	(TransformedTarget, Dummy, GBM, RF, XGB, LGB, Gaussian-Process, Bagging, SGD, Poisson, KNeighbors, RadiusNeighbors, MLP, TheilSen, ExtraTree, DecisionTree) Regressor , (PLS, ARD, Linear) Regression , (Ridge, Elastic-Net, Lasso, Orthogonal Matching Pursuit) CV , LassoLars, Orthogonal Matching Pursuit, ElasticNet, Bayesian Ridge, Lasso, Ridge, LSTMPre, LSTM, LassoLarsIC, LinearSVR, NuSVR	Water level, Rainfall	NSE, KGE, RMSE	Best-performing methods were LSTM, LGBMRegressor, and NuSVR
(Al-Rawas et al. 2024)	DRF, GBM, stacked ensemble learning, GLM, DNN, XRT, SWAT	Specific Humidity, Air Temperature, and Rainfall flux	RMSLE, MSE, RMSE, MAE, MRD	The study reported that the Stacked Ensemble model achieved the best performance for rainfall downscaling under CMIP6 climate scenarios
(Oddo et al. 2024)	LSTM, ConvLSTM	1-h accumulated rainfall data, Soil moisture data, Base reflectivity	RMSE	The ConvLSTM network effectively captured specific spatio-temporal landscape dynamics of a flash flood-prone catchment
(Saint-Fleur et al. 2023)	Deep ANN, MLP	Rainfall, Runoff	NSE, C_p , PPD, SPPD, PD	The proposed deep model improved flash flood peak estimation, and robust during extreme rainfall events.
(Hasnaoui et al. 2025)	RF, CA–Markov, CNN	Elevation, SPI, TWI, Rainfall, SM, AVP, NDVI, Runoff, Distance from River, LULC, Geology	Kno, Klocation, Kstandard	The results reveal a clear link between evolving land use patterns and increased risk of flash floods, especially in urban areas.
(Zhao et al. 2025)	CNN-LSTM-MA, Xinnanjiang (XAJ) model	Rainfall, Discharge, API, RP	HR, FAR, ED, CSI, and Accuracy (ACC)	The proposed CLMA-based probabilistic model outperformed the deterministic model in providing early warnings
(Jhong et al. 2024)	LSTM, LSTM-GA	Water Level, Rainfall	RMSE, R^2 , NSE	Integration of GA improved the performance of LSTM model
(An and Ouyang 2025)	Rainfall-runoff hydrodynamic model, LSTM	Rainfall, Soil moisture	RMSE, NSE	The hybrid framework significantly enhances computational efficiency by enabling the surrogate model to predict runoff dynamics.

satellites have a temporal resolution of essentially 1 day and a spatial resolution of less than 0.25° . Consequently, the aforementioned items are unsuitable for our case study region, since their spatial resolution is excessively coarse for a smaller basin, which may result in loss of precipitation spatial distribution data at low watershed resolutions. In the event of a flash flood, the anticipated leading time can guarantee mass transfer, which is crucial for flood protection (Saharia et al. 2021). The forecast lead time, which includes time for notification and decision making as well as time for action and mitigation, is the amount of time that remains after rainfall began and the danger of a flash flood was identified, until threshold is crossed (Pingel et al. 2005). In small-sized watersheds, the effective lead time of flash flood warnings is minutes to hours (Ming et al. 2020). The main focus of this divided research is how to keep flood prediction accuracy within a reasonable relative forecast time. As a result, we will use forecast lead time of flash floods as one of the most important indicators to confirm the EF5 model's functionality in the case study that follows. Over the last few decades, a lot of research has been performed on the use of distributed runoff prediction models for flood forecasting (Chao et al. 2019). The topic of flash flood forecast modeling and interpolation of precipitation data as input to flood models has been discussed in a number of studies (Foehn et al. 2018; Jiang et al. 2019; Saksena et al. 2020). However, there hasn't been a study that focuses particularly on precipitation input from different interpolation techniques in runoff prediction models with adaptive framework applications, despite this being a significant area of research with unique opportunities and challenges. This study's main goal is to develop and verify an adaptive ensemble framework employing different rainfall interpolation techniques for flood forecasting in the Kendrapara district, and to simulate hourly discharge and forecast it to better understand the behavior of floods in the region.

2 Study Area

Kendrapara District is situated in the north eastern coastal region of the Indian state of Odisha, with geographical coordinates approximately between 20.167°N – 20.750°N latitude and 86.167°E to 87.116°E longitude (Fig. 1). Bay of Bengal borders it to the east, the Bhadrak District to the north, the Jagatsinghpur District to the south, and the Cuttack District to the west. It covers an area of approximately 2774 square kilometers. Flood forecast will be a significant activity in Kendrapara, which is always under threat of inundation, either from the inland or the seaboard. This is based on feeding data from strategically located rain gauge centers and on constant monitoring of the flow of river, which would serve as an important tool in preventive catastrophe management. The anticipation is based on analyzing rainfall patterns from previous years and river flow rates, then predicting in advance the possibility and severity of flooding disasters. They can also issue a warning in advance to the areas most likely to be hardest hit. For such a warning system, it will not mean facilitating evacuation for at risk area communities; they will be allowed to put in place preventive measures mostly for embankment strengthening and provision of emergency response teams, to minimize the impact of the flood. The methodological flow chart of this study is presented in Fig. 2.

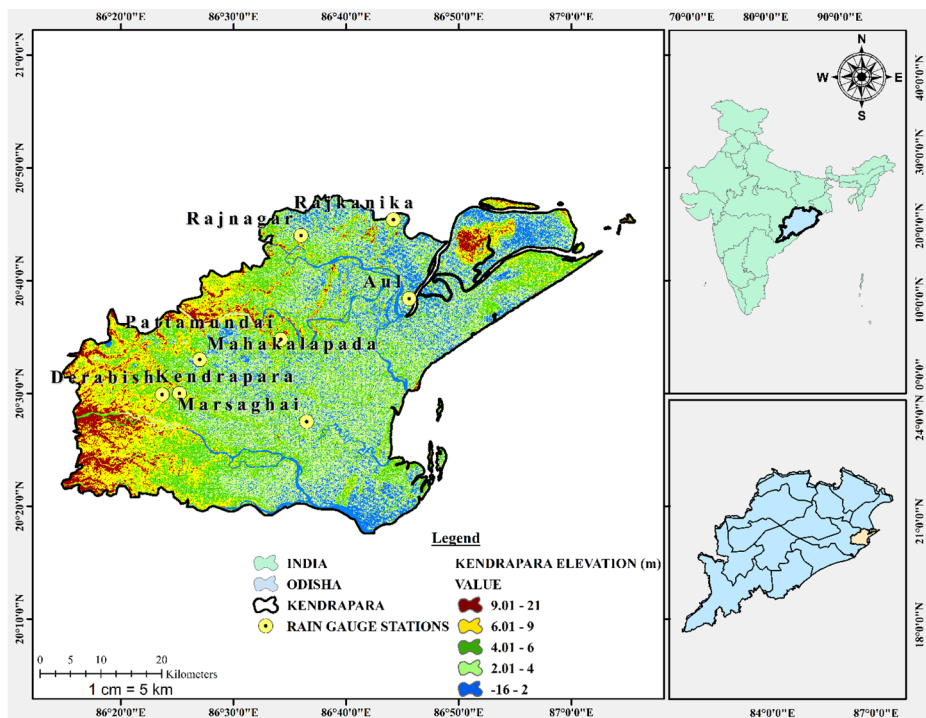


Fig. 1 Geographical location of Kendrapara District in Odisha, India

3 Methodology

3.1 EF-5 Model

The EF5 is a hydrologic modelling platform applied for research and operational hydrological forecasting, in hydrology and general schematization. The EF5 model couples a set of water balance schemes with automatic calibration techniques and kinematic and linear reservoir wave routing as the river routing scheme implementations. The parameters of the EF5 model were calibrated using the various evolution adaptive Metropolis algorithms developed by Vrugt et al. (2008). DREAM is a global parameter optimisation approach using multi chain Monte Carlo simulation in order to get ideal objective values. Based on the paper (Flamig et al. 2020), any water balance model can be made to meet basic specifications and be implemented in EF5.

Other routing options in EF5 include a lumped routing model, often conceptualised as a kinematic wave approximation, and a series of linear reservoirs based on Saint Venant equations for 1D open channel flow. These two concepts are further detailed by Flamig et al. (2020) in their developer documentation. The EF5 model parameters also fall into two classes: KW parameters, which are related to water movement from one cell to another, and CREST parameters, which control the water balance. The parameters are as shown in Table 2. Further calibration of the EF5 model parameters was based on the application of the Evolutionary Adaptive Metropolis (DREAM) technique.

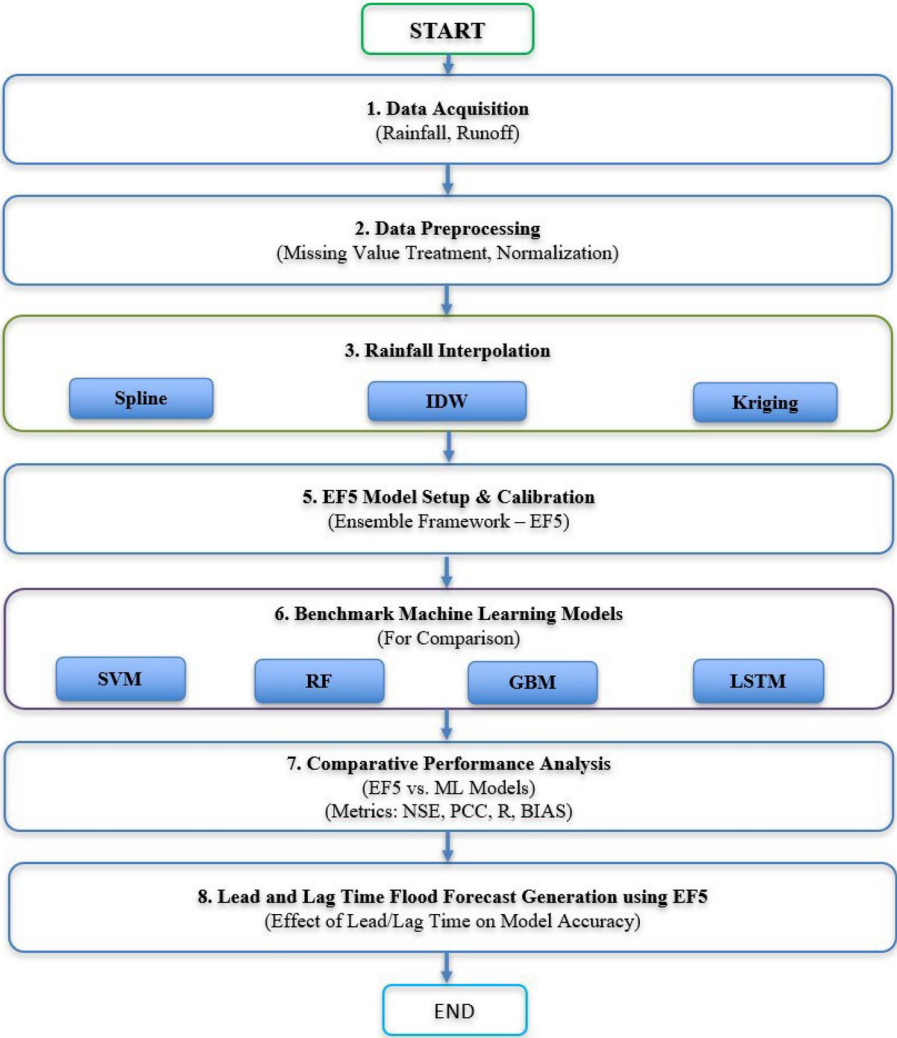


Fig. 2 Flowchart of proposed methodology

Table 2 Parameter values for CREST water balance model and kinematic wave routing model

CREST	W_m	0–2.5	KW	$UNDER$	0.0001–0.99
	F_c	0–3		$LEAKI$	0–130
	b	0–300		ISU	0–350
	I_m	0–3		$Alpha$	0–130
	K_e	0–130		$Beta$	0–130
	Iwu	5–74		$Alpha0$	0–130

3.2 SVM

SVM model is based on principle of structural risk minimization. It has proved to be an efficient and robust algorithm for data analysis, equation fitting, forecasting hydrological parameters, and so on. SVM possess special benefits in solving small sample, high dimensional and non linear pattern recognition problems. SVM can be used in both regression and classification problems.

For classification and regression problems, the primal problem is given Eqs. (1) and (2):

$$\min \frac{1}{2} \|\omega\|^2 \text{ s.t. } y_i (\omega \cdot x_i + b) \geq 1, i = 1, \dots, l \quad (1)$$

Where l is the number of training vectors.

$$\left\{ \begin{array}{l} \min \frac{1}{2} \|\omega\|^2 + C \sum_{i=1}^l (\xi_i + \xi_i^*) \\ \text{s.t.}, f(x_i) - y_i \leq \xi_i^* + \epsilon, i = 1, \dots, l \\ \text{and } y_i - f(x_i) \leq \xi_i + \epsilon, i = 1, \dots, l \\ \text{and } \xi_i \text{ and } \xi_i^* \geq 0, i = 1, \dots, l \end{array} \right. \quad (2)$$

Where, C capacity parameter cost, if error is larger than ϵ , ξ_i and ξ_i^* regulate the degree to which sample points are penalised.

3.3 Random Forest

Among many supervised ML algorithms, RF can be efficiently employed for classification and regression. It comprises several Decision Trees (DTs) and is categorized as an ensemble classifier. Simply stating RF is a classifier built utilising a chain of tree structured classifiers $[h(x, \varnothing_k), k = 1, \dots, l]$, where $(\varnothing_k)$ are independent, equally distributed, or dispersed arbitrary vectors. For the leading prevailing class at input x , each tree casts a single vote. RF can improve prediction as a set of DTs along with a solitary DT. Each variable's importance to the decision is provided by RF. A model is created from the training dataset using DT by learning decision rules for estimating or predicting value or class of target variable.

3.4 Gradient Boosting Machines (GBM)

Another machine learning approach is Gradient Boosting (GB). The concept of developing several classifiers and averaging their results to determine the best one is the foundation of both AdaBoost and GB classifiers. With this paradigm, trees are constructed in a sequential fashion, with each new tree fixing the mistakes of its predecessors. In order to train a decision tree, the algorithm first gives each observation equal weight. Following the evaluation of the first tree, observations that are challenging to classify are given higher weights, while those that are simpler to classify are given lower weights. Consequently, these weighted values serve as the foundation for the second tree. For a fixed number of repetitions, this

procedure is repeated. Subsequent trees assist in classifying observations that were not well classified by earlier trees. In this method, trees are grown by minimizing a cost function. This function is defined as follows:

$$L = \left\langle \sqrt{(Y - Y_{fid})^2} \right\rangle \quad (3)$$

In the above relation, Y represents the predicted model output, and Y_{fid} denotes the actual value of the class to which the input data belongs.

3.5 LSTM

An input gate, a forget gate, and an output gate are included in the architecture of an LSTM. The fading gradients problem is overcome by these elements, and the decision is made on whether to store or yield the input and output information. The state of the memory cell can be altered or blocked by the input gate using an input signal. The state of the memory cell can be allowed to reach the output of the memory cell, or prevented from doing so, with the help of the output gate. The forget gate can allow the memory cell to retain or discard information in its subsequent step. The internal calculation process of the LSTM memory cell is shown in Eqs. (4)–(9) below:

$$f_t = \sigma(W_f X_t + U_f h_{t-1} + b_f) \quad (4)$$

$$i_t = \sigma(W_i X_t + U_i h_{t-1} + b_i) \quad (5)$$

$$\tilde{C}_t = \tanh(W_c X_t + U_c h_{t-1} + b_c) \quad (6)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (7)$$

$$o_t = \sigma(W_o X_t + U_o h_{t-1} + b_o) \quad (8)$$

$$h_t = o_t * \tanh(C_t) \quad (9)$$

where σ indicates the sigmoid function; W_f, W_i, W_c, W_o and U_f, U_i, U_c, U_o represent the weight matrices; X_t signifies current input vector; h_{t-1} and h_t represents output from preceding unit and current unit, b_f, b_i, b_c, b_o stands for the bias vectors, and o_t refers to output gate. The variable $\tilde{C}_t$ utilises $\tanh$ function for computing the pass through weight. Input gate (i_t) decides whether new information is to be discarded or accepted. The newly updated memory (C_t) is the output, C_{t-1} signifies the memory from previous unit. With values ranging between (0, 1), the sigmoid is utilised for expressing switch state of the gate. $\tanh()$ is utilised to update cell state and hidden state with values in the range $(-1, 1)$,

3.6 Data Processing for Interpolation Techniques

Pre processing of information data related to the IDW, Kriging, and Spline methods of rainfall interpolation was acquired from eight strategically placed rain gauge stations in the

Kendrapara region. Such a spatial distribution of rainfall amounts can further enhance the accuracy of runoff estimates by employing methods such as Inverse Distance Weighting, Kriging, and Spline. The data were acquired over three different time periods: from 13 to 16 September 2021, 18 to 21 August 2022, and 2 to 5 August 2023. These periods were chosen because they were followed by heavy rainfall and flooding; hence, this would be an extreme dataset to evaluate the performance of interpolation techniques. The time series plot of the input data for 2021, 2022, and 2023

is shown in Fig. S1 (a), (b), and (c), respectively. The simple and efficient IDW procedure assigns more weight to close stations based on the theoretical premise that closer stations are more differentiable rainfall patterns. Kriging, in contrast, only considers distance but also spatial autocorrelation among data points, offering a more statistically robust estimation. The basic information of data series is provided in Table 3.

The train test split strategy was adopted based on several previous studies conducted across different hydrological modelling aspect (Popa et al. 2019; Nguyen et al. 2024; Hameed et al. 2025; Liu et al. 2026). Among different split values (60:40; 70:30; 80:20; 90:10), the authors selected 80:20 split, i.e., 80% of total data was used during training phase, and remaining 20% data was used for testing phase (Asif et al. 2026). The authors considered four days of hourly data (96 h) from three different years, i.e., 2021 (13/08–16/08), 2022 (18/08–21/08), and 2023 (02/08–05/08). The first 72 h of data were used for training, while the rest 24 h of data was used for model testing. Normalisation of the data is done using the formula depicted in Eq. (10) to achieve normalized values in each column offering dimensionless data between 0 and 1.

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (10)$$

Where X_{max} and X_{min} - maximum and minimum values of each input; X_{norm} - normalised value.

Our results show the advantages and disadvantages of all methods for Kendrapara's rainfall data and for estimating surface runoff. The interpolations results are provided in Table S1. While IDW appears preferable in terms of simplicity and computational cost, it may not fully reflect spatial variability when rainfall patterns are strongly affected by local geographic features, resulting in location referenced runoff estimates that may be less accurate. As Kriging is based on geostatistics, it provides more accurate and reliable

Table 3 Parameters of dataset include length of sequence (L), standard deviations (S_x), average values, maximum value (X_{max}), minimum value (X_{min}), bytes of every input rainfall files, and pixel of all rainfall raster

DATASET	Statistical parameters							
Discharge (m ³ /s)	L	S _x	X _{max}	X _{min}	Mean	Median		
2021–1308–1608-hourly	95	6.808	27.011	1.067	13.283	11.646		
2022–1808–2108-hourly	95	6.003	23.815	0.931	11.712	10.269		
2023–0208–0508-hourly	95	6.907	26.719	0.806	12.658	10.82		
Rainfall(mm)	L	S _x	X _{max}	X _{min}	Mean	Median	Bytes	Pixel
2021–1308–1408-hourly	95	6.875	27.457	0.629	10.803	8.995	08.80G	1.16×10^{-1}
2022–1808–2108-hourly	95	6.035	26.91	0.12	10.539	10.36	08.14G	1.16×10^{-1}
2023–0208–0508-hourly	95	13.477	50.62	0	16.665	12.69	07.80G	1.16×10^{-1}

interpolation results if the spatial structure of the data is well understood, consequently providing better inputs for runoff models. Nonetheless, Kriging is computationally heavier and demands a better understanding of the latent spatial processes. To achieve similar outcomes in data collection, researchers worked with strategically positioned rain gauge stations instead of random locationbased data. Station data ensure that measurements are taken at fixed, scientifically chosen locations, thereby providing a systematic approach to monitor rainfall and its spatial variability accurately. In this regard, using the selected periods, the performance of these interpolation methods will be analyzed to find out which method better fits the estimation of runoff in Kendrapara. This information shall help stakeholders in local water resource management and planning by providing more accurate and precise runoff predictions and making decisions based on such rainfall events.

It uses diversified interpolation to generate precipitation input files that belong to the same flood episodes as the EF5 model, with identical time steps and a common timeline. Three interpolation methods were applied to obtain various rainfall distribution raster with the EF5 model. The EF5 model's parameters, as already mentioned, were calibrated automatically with the DREAM program (Karamouz et al. 2014).

3.7 Simulation Verification

Performance of models is evaluated using statistical characteristics of the outputs. The accuracy of simulations is measured utilizing PCC, expressed by:

$$PCC = \sqrt{\text{Cov}(Q_p, Q_0) / (\text{Var}(Q_p) \text{Var}(Q_0))} \quad (11)$$

here, Q_p simulated discharge, Q_0 actual discharge. PCC varies from -1 to 1 ; 1 indicates the strongest association. Disparity of two series is verified using normalised bias of flood simulation, and is calculated as:

$$BIAS = \sum_{t=1}^N \left(\frac{Q_p - Q}{Q} \right) \quad (12)$$

where, N - number of observations. BIAS ranges from -1 to ∞ , with 0 being least biased. However, other study indicated that utilizing only PCC and BIAS is an unsatisfactory approach for evaluating hydrologic models. As suggested by (Ritter and Muñoz-Carpena 2013), using a mix of graphical findings, along with absolute value error indicators (RMSE), and normalised goodness of fit indicators. Finally, it employed the NSE and the RMSE to express the divergence between forecasts and observations, as shown below in Eqs. 13 and 14:

$$RMSE = \sqrt{\frac{1}{N} \sum_{t=1}^N (Q_P - Q_0)^2} \quad (13)$$

$$NSE = 1 - \frac{\sum_{i=1}^N (Q_p - Q_0)^2}{\sum_{i=1}^N \left(Q_p - \bar{Q}_0 \right)^2} \quad (14)$$

The RMSE values range from 0 - ∞ , whereas NSE ranges from minus ∞ to 1. Both 0 for RMSE and 1 for NSE indicate that simulated values exactly match observed values.

4 Results and Discussion

4.1 Discharge Forecast Simulation Performance of EF5

Longterm prediction using observed data is typically more perplexing than short-term forecast simulations due to improved uncertainty over extended periods (Jiang et al. 2019). The simulated vs. observed discharge for three consecutive years (2021, 2022, and 2023) is presented in Fig. 3. Daily precipitation raster data for the entire years of 2021, 2022, and 2023, obtained through three different interpolation methods, were input into the EF5 model to generate runoff forecast simulations for the Kendrapara district. The EF5 model parameters were automatically calibrated utilising the DREAM algorithm (Karamouz et al. 2014). Figure 4 shows a comparison amid observed discharge values and forecasted simulation values generated by the different interpolation methods. This comparison helps in assessing

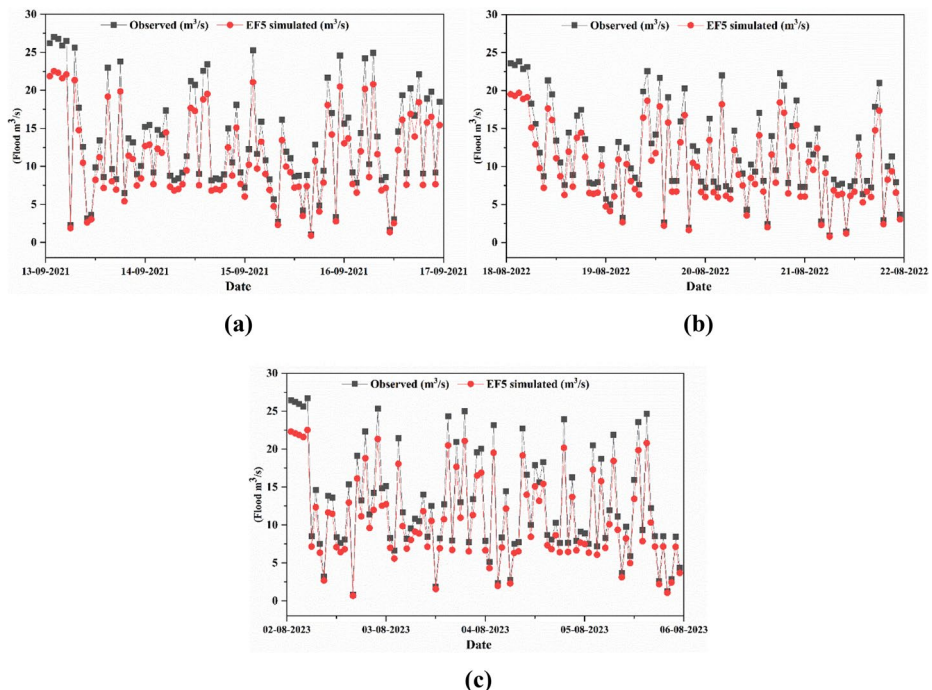


Fig. 3 Observed vs. Simulated Discharge for (a) 2021, (b) 2022, and (c) 2023

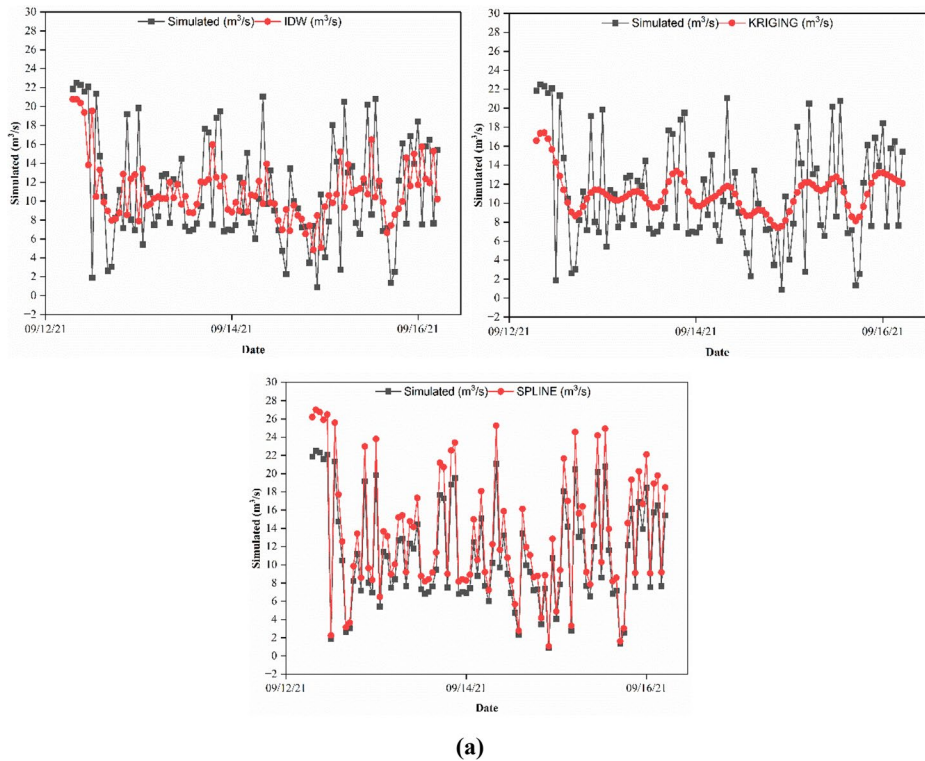


Fig. 4 Interpolation vs. Simulated Discharge Validation (IDW, Kriging and Spline) for (a) 2021, (b) 2022 and (c) 2023

the performance and accuracy of different interpolation methods for predicting flood events. Integrating high resolution MRMS data, radaronly precipitation files, and potential evapo-transpiration files, the study intends to refine a flooding failsafe model that can be used in flood risk management.

4.2 Prediction Performance of Machine Learning Models

The training and testing model performance metrics shown in Table S2 represent a comprehensive evaluation of SVM, RF, LSTM, and GBM models to assess their ability to predict flash flood in the Kendrapara region. It can be observed from the results shown in Table S2 that each model showed its own strengths in different areas of predictive accuracy. With a PCC of 0.9332, 0.909, and 0.9184 for 2021, 2022, and 2023, GBM exhibited excellent performance with accurate predictions. These values pointed up the model's ability to capture the underlying trends in the flow data. RF outperformed LSTM and SVM, achieving the higher values for PCC (0.9042, 0.7635, 0.8213) and NSE (0.8033, 0.5677, 0.6627) respectively for years 2021, 2022, and 2023. Its robustness and the strong correlation between predicted and observed flood values are represented in its high R^2 values. However, in testing phase, LSTM showed better results than all other models. The graphical time-series plot of the simulated results from the different models are shown

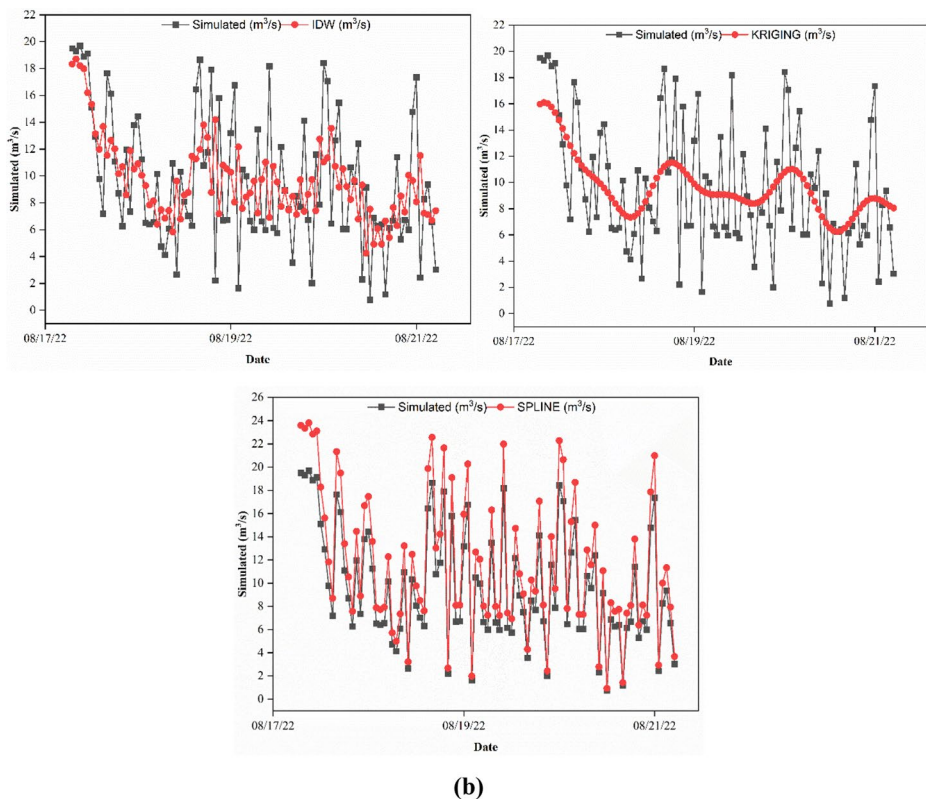


Figure 4 (continued)

in Figs. S2–S5. Fig. S2 (a), (b), and (c) show the time-series plot of the observed vs. SVM predicted flood values for 2021, 2022, and 2023, respectively. Similarly observed vs. RF-predicted, LSTM-predicted, and GBM-predicted flood values are demonstrated in Figs. S3 (a), (b), (c), S4 (a), (b), (c), and S5 (a), (b), (c), respectively for the years 2021, 2022, and 2023. The lead time forecasts of the ML models (SVM, RF, LSTM, GBM) are provided in Tables S4–S7.

4.3 Prediction Performance of EF5 for Leading/Lagging Forecast Time

Forecasting flash floods, which is associated with early warning and allowing time for safe transfer, depends heavily on prediction leading time. The lead time components have been described in earlier research (Krzyszhanovskaya et al. 2011). A tradeoff was observed amid longer lead times and worse reliability, pointing out that the uncertainty increases with the early forecasting of a flood hazard (Goodarzi et al. 2019). Flood forecasting necessitates at least an hour's notice to ensure individuals have enough time to evacuate (Pingel et al. 2005). The validation results for leading forecast (Lead+1 H, Lead+3 H, Lead+6 H, Lead+10 H) simulation are given in Table S3. For the two simulation scenarios with a singular flood peak, it closely examines EF5 model's leading time prediction performance and computes verification statistics with the shift in the forecast lead time. To ensure the safety

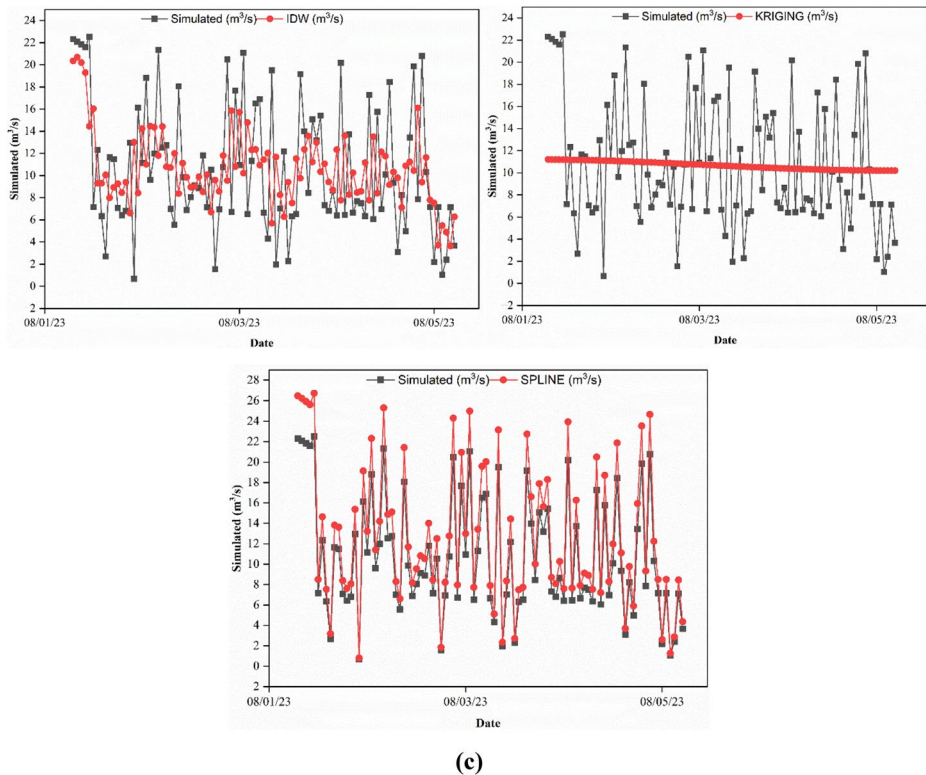


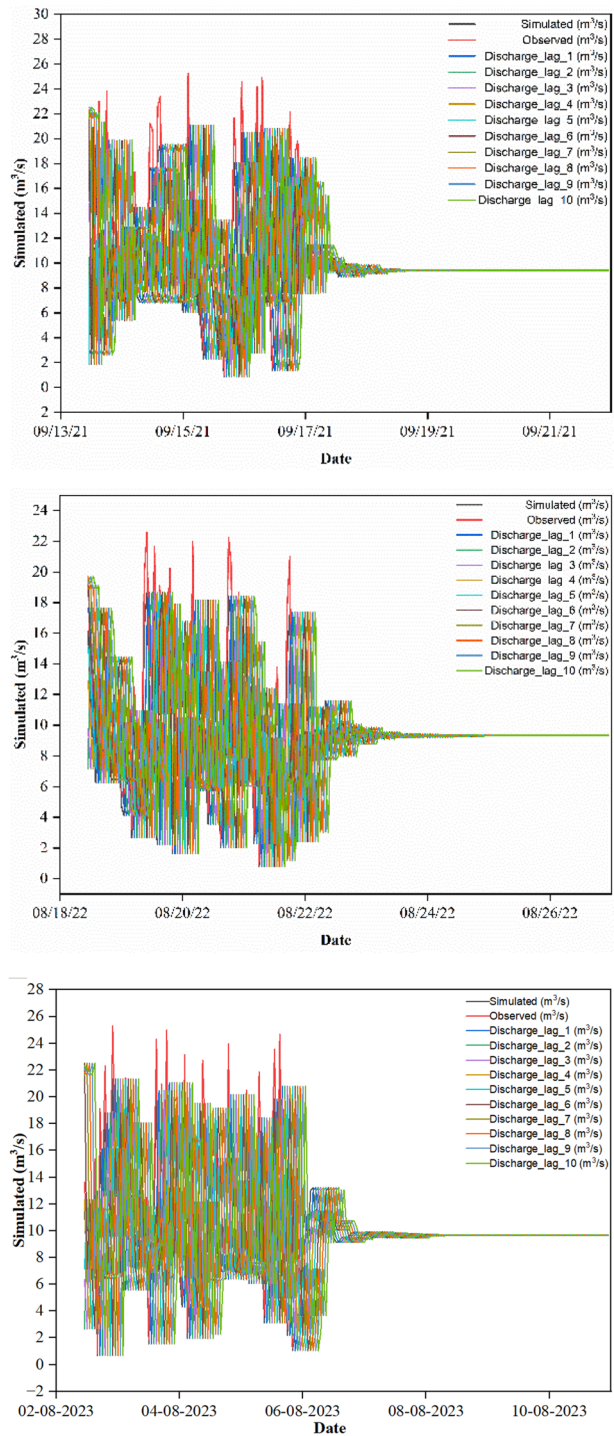
Figure 4 (continued)

and practicality of flash flood prevention, the rainfall input file is deleted many hours before to standard time point, which is determined by taking the runoff process's flood peak timing. After being calculated as indicated, PCC, NSE, and BIAS all exhibit a declining trend with an increase in the flood forecast lead time.

Lag forecast timeseries for the EF5 model during the same years are displayed in a collection of graphs (Fig. 5). Simulated discharge values in the 2021 graph oscillate between 10 and 30 m³/s from 09/13/21 to 09/21/21. Observed values exhibit a similar, but more erratic, pattern. The displayed discharge values have lag durations ranging from one to ten hours. Shorter lags (such as lag 1) closely correspond to the actual values, whereas longer lags exhibit significant divergence, which denotes a decline in accuracy over time. The simulated values in the 2022 graph span the dates 08/18/22 to 08/26/22, and they range from roughly 10 m³/s to 24 m³/s.

The recorded discharge values have peaks, particularly around 08/22/22. Predictive values for higher lags (greater than lag 2) show greater variability and divergence, reflecting the patterns in these lead forecasts. In contrast, shorter lag times (specifically lag 1 and lag 2) align better with reality. The graph for 2023 shows simulated discharge values from August 02, 2023 to August 10, 2023, varying from about 8 m³/s to around 28 m³/s, but there are large peaks in recorded discharge, with the highest between 08-05-2023 and 08-08-23. This is consistent with data from earlier years, which showed that forecast accuracy fell as

Fig. 5 Lag forecast time-series for EF5 model



lag time increased; short lag forecasts remain close to observed values, while longerlag forecasts drift away. The validation for leading forecasting simulation by EF5 Model is provided in Table 4. The table shows PCC, NSE, and BIAS values for different lead times.

This graph set focuses on the EF5 model's lead forecast time series for 2021, 2022, and 2023 (Fig. 6). The simulated discharge values in the 2021 graph exhibit considerable fluctuations between 10 m³/s and 30 m³/s from 09/13/21 to 09/21/21. There are differences between the model and actual observations, as evidenced by the observed discharge, which largely follows this trend, although with greater fluctuations. Plotting lead times ranging from 1 to 10 h demonstrates how predictions change over time. Notably, larger lead times exhibit greater divergence, suggesting that shortterm forecasts are more dependable, but the lead-1 forecast remains closer to the observed values. The pattern is similar in the 2022 graph, where the simulated values range from 10 m³/s to 24 m³/s from August 18, 22 to August 26, 22. More dramatic fluctuations are seen in the recorded discharge, especially around 08/22/22, when it rises noticeably above 20 m³/s. Once more, shorter lead periods (such as lead 1) yield projections that are more in line with the observed values. In contrast, longer lead times exhibit a pattern comparable to 2021, but with usually lower discharge values.

The plot for 2023 shows the simulated discharge test for the period from (08/02/23) to (08/10/23), with values ranging from about 8 m³/s to 28 m³/s. The observed values also fluctuate in the same way, having two peaks around the dates (08/05/23) and (08/08/23). As lead time increases, disagreements also grow, but on shorter lead times like lead 1 and lead 2, predictions are closer to the observed data, confirming a trend from previous years: short-term forecasts tend to be accurate. In summary, as illustrated across all three sets of graphs, the challenge is maintaining forecast accuracy at longer lead and lag intervals shorter lead times are typically more accurate than longer ones. Trends, so far consistent over time, point toward persistent problems with forecast divergence another important consideration when refining prediction models.

The group of charts, as shown in Fig. S6, uses NSE, PBIAS, and PCC to evaluate and show how well things worked in 2021, 2022, and 2023 across different time lengths. The NSE (Nash Sutcliffe Efficiency) graph shows that the models work quite differently depending on how far ahead they try to predict. In 2021, the NSE starts at 0.5 when predicting right now, goes up and down, and ends up at 0 when predicting 10 h ahead. In 2022, you can see the NSE drops quite a bit, reaching -0.5 when predicting about six hours ahead, but then it starts to go back up. On the other hand, in 2023, the NSE stayed pretty steady, except for a dip when predicting 4 to 6 h ahead, suggesting the model struggled to get it right during those periods. The PBIAS (Percent Bias) graph shows how much model bias changes over lead times. For 2021, PBIAS starts at about 0.6 at 0 h and slowly falls, hitting about -0.1 by 10 h. The 2022 data has more bias beginning at 0.5 and sharply dropping to -0.4 around 6 h of lead time. This points to a big over or underestimation by the model during this period. In 2023, PBIAS kicks off at a level close to 2021 and remains steadier, but it dips at the 6 to 8 h mark compared to 2022, hinting at less steady bias. The Pearson Correlation Coefficient (PCC) graph measures how closely observed and expected values align. In 2021, PCC starts at 1.0 at 0 h, showing a perfect match. But by 10 h, it drops to about 0.2, losing accuracy. The 2022 data falls even faster, dipping below 0.2 after 6 h. On the other hand, 2023 doesn't drop as much. It levels off at 0.3 after 10 h. This indicates that the new model retains its accuracy longer than in previous years.

Table 4 Validation for leading forecast simulation by EF5 model

Model	13/09/2021 00:00–16/09/2021 23:00				18/08/2022 00:00–21/08/2022 23:00				02/08/2023 00:00–04/08/2023 23:00			
	Calibration				Validation							
EF5	R ²	PCC	NSE	BIAS	R ²	PCC	NSE	BIAS	R ²	PCC	NSE	BIAS
	0.9291	0.9638	0.9187	0.0574	0.7964	0.8924	0.7882	0.0812	0.7964	0.8924	0.7882	0.0812
EF5	R ²	PCC	NSE	BIAS	R ²	PCC	NSE	BIAS	R ²	PCC	NSE	BIAS
	0.9473	0.9732	0.9402	0.0059	0.8625	0.9287	0.8562	0.0725	0.8625	0.9287	0.8562	0.0725
EF5	R ²	PCC	NSE	BIAS	R ²	PCC	NSE	BIAS	R ²	PCC	NSE	BIAS
	0.9158	0.9569	0.9075	0.0862	0.7593	0.8713	0.7504	0.0984	0.7593	0.8713	0.7504	0.0984

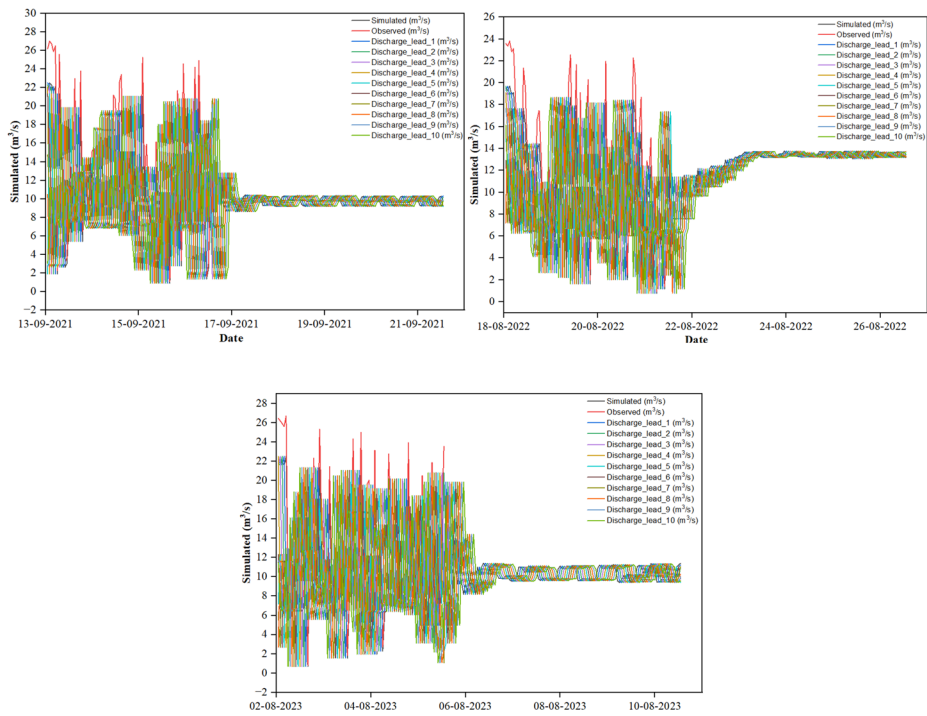


Fig. 6 Lead forecast time-series for EF5 model

4.4 Discussion

Because of the unique geographical and climatic conditions in Kendrapara District, which are prone to frequent flooding, this study aims to address flash flood prediction problems. Utilizing the EF5 model, which was calibrated to local conditions in Kendrapara, will help officials and stakeholders obtain timely, accurate flood forecasts. It enables them to function by establishing warning systems, planning evacuations, and allocating resources to mitigate the effects of flooding on communities, infrastructure, and livelihoods within the district. Besides, IDW has been found to perform best amongst various rainfall mapping methods, and selection of such techniques is crucial for estimating precipitation in Kendrapara District. It provides a clear view of rainfall distribution and maps precipitation values using IDW. This improves flood forecast accuracy, preparing and responding to the region in time.

These findings bring immense value to the literature and inform flood management and decision making processes in Kendrapara District. Application of the EF5 model and optimal mapping of rainfall help the authorities enhance flood prediction and mitigation capacity, saving lives, property, and infrastructure in this vulnerable area. In this study, rainfall mapping was assessed using three approaches across three flood events using the EF5 prediction model. It tested several mapping methodologies and created a rainfall map optimized for the input of the EF5. A forecast for long-term data is generally more challenging than a short-term forecast because uncertainty increases with time. Daily rainfall maps for the period 2021–2023, obtained using three different mapping techniques will be used as inputs to the EF5 model to predict runoff.

5 Conclusion

This study has presented useful information on the flash flood forecast in Kendrapara, a coastal district with distinctive water patterns. It shows how the EF5 model may be helpful in forecasting floods in this sensitive district. The EF5 model uses local maps for developing a realizable runoff forecast system, which is cross validated using data from water stations. The results of EF5 model were compared with four benchmark machine learning models, namely RF, SVM, LSTM, and GBM. Comparison of results showed that the proposed EF5 model for flash flood forecasting showed better performance than the ML models for all scenarios. The validation shows that the model produces realistic, practical short-term flood forecasts that indicate peak water flow and timing. In another study, by comparing rainfall mapping techniques such as IDW, Kriging, and Spline, rainfall could be mapped to give fine flood predictions using spline maps and supply its distribution in a very precise manner. This confirms that the model is versatile and reliable for predicting floods in this region, as accurate flood prediction models are developed using local map data. For the EF5 model, except for the Kriging Mapping Method, performance was checked against simulated and real survey runoff data, yielding very high PCC and NSE scores. This reflects a satisfactory level of fit between the modeled and observed records and attests to the suitability of the model for predicting short-term floods in Kendrapara. However, we observe that the performance of our model does not hold well while forecasting for longer durations. The cryptic nature of these dynamics has emphasized the need to develop flood prediction techniques specific to regions of the coast, such as Kendrapara. This has impacts on effective risk reduction, as a reasonable time window for flood forecasting needs to be achieved (which was one of the prime findings of this study). The inference is that this study finds the relevance of developing an integrated flood forecasting system tailored for a coastal district like Kendrapara, and that it would be precise and provide lead time. Further efforts need to be made by the researchers to improve the adaptive ensemble framework, which can further focus attention on increasing efficiency in forecasting. They can do so by integrating new data sources, such as higher-resolution satellite imagery and remote sensing data. It will also be useful to test this framework's performance and adaptability in other flood prone areas with different climates and topography. This strategy will show how effective it is in many situations.

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Declarations

Conflict of interest The authors declare no conflict of interest to any party.

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